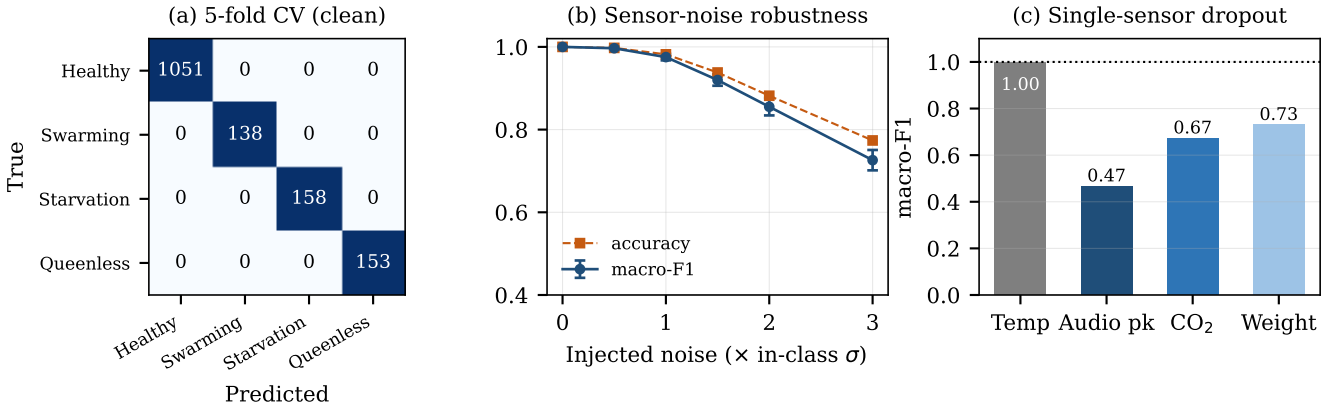
(a) 5-fold CV (clean)
Healthy
Swarming
Starvation
Queenless
True
1051
0
0
0
0
138
0
0
0
0
158
0
0
0
0
153
Healthy
Swarming
Starvation
Queenless
Predicted
(b) Sensor-noise robustness
1.0
0.8
0.6
0.4
accuracy
macro-F1
0
1
2
3
Injected noise (× in-class σ)
(c) Single-sensor dropout
macro-F1
1.0
0.8
0.6
0.4
0.2
0.0
1.00
0.47
0.67
0.73
Temp
Audio pk
CO2
Weight